

Boss Challenge — Paper 26 (Class 7)

Electric Current & its Effects | Simple Equations | Reproduction in Plants | Data Handling

One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. A small device used to break or complete the path of electric current in a circuit is the:
(A) cell
(B) filament
(C) switch
(D) fuse
2. A torch already holds some cells. When 2 more cells are pushed in, it has 5 cells in all. Writing this as $x + 2 = 5$, the number it started with is:
(A) 2
(B) 10
(C) 7
(D) 3
3. Reproduction in which a new plant grows from a part of the parent (root, stem or leaf) without seeds is called:
(A) vegetative propagation
(B) fertilisation
(C) pollination
(D) dispersal
4. Over 5 days a torch was switched on for 3, 5, 2, 4 and 6 hours. The arithmetic mean (average) number of hours per day is:
(A) 20
(B) 4
(C) 5
(D) 6
5. When a magnetic compass needle is brought near a wire carrying current, the needle deflects. This shows that an electric current has a:
(A) chemical effect only
(B) magnetic effect
(C) no effect at all
(D) cooling effect
6. Find the value of the unknown number that makes the statement $x + 9 = 21$ true.
(A) 30
(B) 189
(C) 12
(D) 3

7. The male reproductive part of a flower, made of an anther and a filament, is the:
- (A) petal
 - (B) pistil
 - (C) sepal
 - (D) stamen
8. The arithmetic mean (average) of the three numbers 4, 6 and 8 is:
- (A) 18
 - (B) 6
 - (C) 8
 - (D) 3
9. A coil of insulated wire wound around an iron piece becomes a magnet only while current flows. This is called an:
- (A) permanent magnet
 - (B) fuse
 - (C) electromagnet
 - (D) compass
10. A sunflower head holds the same number of seeds, x . Three identical heads together hold 150 seeds. Using $3x = 150$, the number of seeds in one head is:
- (A) 450
 - (B) 147
 - (C) 153
 - (D) 50
11. The female reproductive part of a flower, made of the stigma, style and ovary, is the:
- (A) sepal
 - (B) stamen
 - (C) pistil (carpel)
 - (D) petal
12. In the data set 2, 3, 3, 5 and 7, the value that occurs most often (the mode) is:
- (A) 4
 - (B) 7
 - (C) 3
 - (D) 5
13. Inside a glowing electric bulb, the thin coiled tungsten wire that becomes white-hot is known as the:
- (A) filament
 - (B) switch
 - (C) fuse
 - (D) terminal

14. What number makes the equation $6x = 54$ correct?

- (A) 9
- (B) 60
- (C) 48
- (D) 324

15. The transfer of pollen grains from the anther of a flower to its stigma is called:

- (A) pollination
- (B) dispersal
- (C) germination
- (D) fertilisation

16. When the numbers 4, 1, 7, 3 and 9 are arranged in order, the middle value - the median - is:

- (A) 7
- (B) 9
- (C) 5
- (D) 4

17. When current passes through a wire, the wire becomes warm. This is called the ____ of electric current.

- (A) magnetic effect
- (B) freezing effect
- (C) heating effect
- (D) lighting effect

18. Solve the equation in which a number divided by 5 gives 7, written as $x / 5 = 7$.

- (A) 35
- (B) 12
- (C) 1.4
- (D) 2

19. When pollen from a flower reaches the stigma of a flower on a DIFFERENT plant, it is called:

- (A) cross-pollination
- (B) fertilisation
- (C) vegetative propagation
- (D) self-pollination

20. The range of the data set 12, 5, 18, 9 and 7 is found by subtracting the lowest value from the highest. The range is:

- (A) 13
- (B) 5
- (C) 23
- (D) 18

21. In a circuit diagram, a single cell is drawn as a long line and a short thick line. The long line stands for the:

- (A) switch
- (B) negative terminal
- (C) filament
- (D) positive terminal

22. Find the value of x in the equation $2x + 1 = 13$.

- (A) 12
- (B) 14
- (C) 6
- (D) 7

23. After fertilisation, the ovary of a flower grows and ripens into a:

- (A) petal
- (B) fruit
- (C) leaf
- (D) root

24. Over 5 days a plant scattered 10, 12, 8, 10 and 10 seeds. The value that occurs most often (the mode) is:

- (A) 10
- (B) 8
- (C) 50
- (D) 12

25. When two or more cells are joined together to give more push to the current, the combination is called a:

- (A) battery
- (B) circuit
- (C) conductor
- (D) switch

26. In a circuit each cell gives a 'push' of 2 units. A battery of x cells gives a total push of 12 units. Using $2x = 12$, the number of cells is:

- (A) 24
- (B) 10
- (C) 6
- (D) 14

27. Ferns and mosses do not make seeds. Instead they reproduce by means of tiny:

- (A) fruits
- (B) seeds
- (C) spores
- (D) tubers

28. In a pictograph one full picture of a battery stands for 5 cells sold. A shelf row shows 4 full battery pictures. The number of cells sold is:

- (A) 20
- (B) 5
- (C) 9
- (D) 4

29. A safety wire that melts and breaks the circuit when too much current flows is the:

- (A) electromagnet
- (B) filament
- (C) switch
- (D) fuse

30. 'Five more than a number is fourteen.' Turning this sentence into the equation $x + 5 = 14$, the number is:

- (A) 19
- (B) 9
- (C) 70
- (D) 2.8

31. Simple water plants such as Spirogyra break into pieces, and each piece grows into a new plant. This is called:

- (A) pollination
- (B) germination
- (C) budding
- (D) fragmentation

32. Tossing one fair coin a single time, what is the chance (probability) that it lands showing a head?

- (A) 0
- (B) 2
- (C) $1/2$
- (D) 1

33. The working of an electric bell depends mainly on the ____ of electric current.

- (A) heating effect
- (B) chemical effect
- (C) cooling effect
- (D) magnetic effect

34. Solve for x: $3x - 4 = 20$.

- (A) 24
- (B) 16
- (C) 8
- (D) 72

35. The brightly coloured part of a flower whose main job is to attract insects for pollination is the:
- (A) sepal
 - (B) anther
 - (C) ovary
 - (D) petal
36. When an ordinary six-faced die is rolled once, the probability of getting a number GREATER than 4 is:
- (A) $\frac{1}{6}$
 - (B) $\frac{1}{3}$
 - (C) $\frac{2}{3}$
 - (D) $\frac{1}{2}$
37. Materials such as copper and iron that let electric current pass through them easily are called:
- (A) insulators
 - (B) switches
 - (C) filaments
 - (D) conductors
38. A ripe pod always bursts into x seeds. Five such pods scatter 35 seeds altogether. Using $5x = 35$, the number of seeds in one pod is:
- (A) 175
 - (B) 7
 - (C) 40
 - (D) 30
39. Light seeds with wing-like or hair-like parts, such as those of the drumstick or maple, are usually dispersed by:
- (A) wind
 - (B) animals
 - (C) explosion
 - (D) water
40. A bar graph is best described as a way of showing data using:
- (A) slices of a circle for each part
 - (B) bars of equal width whose heights stand for the quantities
 - (C) pictures that each stand for a number
 - (D) a single line joining changing points
41. An electric bulb stops glowing because its filament has broken. We say the bulb has:
- (A) become a battery
 - (B) more current flowing through it
 - (C) turned into a conductor
 - (D) fused (its circuit is now open)

42. Which value of the unknown satisfies the equation $4x = 28$?

- (A) 32
- (B) 7
- (C) 112
- (D) 24

43. A coconut can float across the sea and sprout on a distant shore. Its seed is dispersed mainly by:

- (A) wind
- (B) water
- (C) explosion
- (D) animals

44. The median of the four ordered current readings 3, 5, 9 and 11 (in milliamperes) is the average of the two middle values, which is:

- (A) 9
- (B) 8
- (C) 7
- (D) 14

45. A circuit in which the cell, switch and bulbs are all joined one after another in a single loop is a _____ circuit.

- (A) broken
- (B) series
- (C) chemical
- (D) magnetic

46. A series light-line holds x glowing bulbs plus 3 spare bulbs, making 10 bulbs in all. Using $x + 3 = 10$, the number of glowing bulbs is:

- (A) 3
- (B) 13
- (C) 30
- (D) 7

47. Seeds of plants like Xanthium have tiny hooks or spines so that they can be dispersed by:

- (A) clinging to the fur of animals
- (B) floating on still water
- (C) bursting out of a dry pod
- (D) being blown by a gentle breeze

48. An event that is certain to happen, such as the sun rising tomorrow, has a probability of:

- (A) $\frac{1}{2}$
- (B) 0
- (C) 100
- (D) 1

49. Why is the heating element (coil) of an electric heater made of nichrome rather than copper?

- (A) nichrome has high resistance and a very high melting point
- (B) nichrome carries no current at all
- (C) nichrome melts very easily when warm
- (D) copper gives out more heat than nichrome

50. 'Twice a number, decreased by 5, equals 9.' Writing this as $2x - 5 = 9$, the number is:

- (A) 28
- (B) 7
- (C) 2
- (D) 14

51. Why is the dispersal of seeds away from the parent plant useful?

- (A) it stops the seeds from ever growing
- (B) it makes the parent plant grow taller
- (C) it removes the need for sunlight
- (D) it prevents overcrowding and competition for water, sunlight and minerals

52. To find the arithmetic mean of a set of observations, you:

- (A) pick the value that appears most often
- (B) subtract the smallest from the largest
- (C) add up all the observations and divide by how many there are
- (D) pick the middle value once they are ordered

53. Unlike a permanent magnet, a big advantage of an electromagnet is that it:

- (A) is always weaker than a compass needle
- (B) can never be turned off
- (C) can be switched on and off and its strength changed
- (D) works even with no current at all

54. Solve the equation $x - 12 = 0$.

- (A) 12
- (B) -12
- (C) 24
- (D) 0

55. A potato has small 'eyes' that can sprout into new plants. This is an example of vegetative propagation by a:

- (A) flower
- (B) leaf
- (C) stem (tuber)
- (D) seed

56. Four pods held 6, 8, 8 and 10 seeds. The mean (average) number of seeds per pod is:
- (A) 10
 - (B) 8
 - (C) 32
 - (D) 4
57. Adding more cells (joined correctly) to a torch circuit usually makes the bulb glow brighter because it:
- (A) increases the current pushed through the filament
 - (B) turns the current into sound
 - (C) cools the filament down
 - (D) removes the filament from the bulb
58. A flower has 4 times as many pollen-bearing stamens as petals. It has 20 stamens. Using $4x = 20$, the number of petals is:
- (A) 24
 - (B) 16
 - (C) 80
 - (D) 5
59. The leaves of the Bryophyllum plant grow tiny buds along their notched edges. Each bud can grow into a new plant. This shows vegetative propagation by:
- (A) pollen
 - (B) leaves
 - (C) seeds
 - (D) roots
60. An event that can never happen, such as rolling a 7 on an ordinary six-faced die, has a probability of:
- (A) $\frac{1}{6}$
 - (B) 1
 - (C) 0
 - (D) 7
61. Touching a bare live wire is dangerous mainly because:
- (A) the wire turns into a magnet in your hand
 - (B) the body blocks all current completely
 - (C) the human body conducts current, which can give a harmful shock
 - (D) live wires are always very cold
62. Solve for x when half of a number, increased by 3, equals 8, written as $(x / 2) + 3 = 8$.
- (A) 22
 - (B) 5
 - (C) 10
 - (D) 11

63. In sexual reproduction in plants, a new individual begins when:
- (A) a leaf bud falls and sprouts
 - (B) a male gamete (from pollen) fuses with a female gamete (egg)
 - (C) a piece of stem grows roots on its own
 - (D) a spore lands on wet soil
64. The most suitable average to answer 'which shoe size is the most common in the class?' is the:
- (A) mode
 - (B) mean
 - (C) range
 - (D) median
65. The filament of an electric bulb is usually made of tungsten because tungsten:
- (A) stays completely cold when current flows
 - (B) has a very high melting point and glows white-hot without melting
 - (C) melts as soon as it warms up
 - (D) does not allow any current through
66. What value of x makes the equation $x + 4 = 4$ true?
- (A) 0
 - (B) 1
 - (C) 8
 - (D) 4
67. The single cell formed when the male and female gametes fuse during fertilisation is called the:
- (A) pollen
 - (B) zygote
 - (C) stigma
 - (D) petal
68. For which task would a double bar graph (two bars per category) be the most useful kind of chart?
- (A) compare two sets of data side by side for each category
 - (B) show how one quantity changes smoothly over time
 - (C) display only a single set of values
 - (D) show the parts of a single whole
69. A modern safety device that, like a fuse, cuts off the supply on overload but can simply be switched back on is the:
- (A) ordinary fuse
 - (B) filament
 - (C) battery
 - (D) MCB (miniature circuit breaker)

70. Solve the equation $5x + 2 = 2x + 17$, which has the unknown on both sides.

- (A) 15
- (B) 3
- (C) 5
- (D) 19

71. A flower that has both stamens (male) and a pistil (female) in the same flower is called a:

- (A) bisexual flower
- (B) spore
- (C) unisexual flower
- (D) seed

72. Five daily current readings were 4, 4, 6, 5 and 6 milliamperes. Their mean (average) is:

- (A) 4
- (B) 6
- (C) 5
- (D) 25

73. The magnetic field produced by a current-carrying coil becomes stronger when you:

- (A) paint the coil a darker colour
- (B) reduce the current to zero
- (C) increase the current or add more turns to the coil
- (D) remove all the turns of the coil

74. The number of seeds in a fruit is 3 more than twice the number of carpels. The fruit has 11 seeds. Using $2x + 3 = 11$, the number of carpels is:

- (A) 7
- (B) 14
- (C) 8
- (D) 4

75. Gardeners grow new money plants and rose plants quickly by planting pieces cut from a healthy stem. This artificial method is called growing from:

- (A) cuttings
- (B) spores
- (C) pollen
- (D) seeds

76. When a fair die is rolled once, the probability of getting an EVEN number is:

- (A) $\frac{1}{6}$
- (B) $\frac{1}{3}$
- (C) $\frac{2}{3}$
- (D) $\frac{1}{2}$

77. Appliances such as an electric iron, a toaster and a room heater all work mainly using the:

- (A) cooling effect of electric current
- (B) chemical effect of electric current
- (C) magnetic effect of electric current
- (D) heating effect of electric current

78. Solve the equation $10 - x = 3$.

- (A) 7
- (B) -7
- (C) 13
- (D) 30

79. Ferns and mosses spread very successfully by spores mainly because spores are:

- (A) made only by flowering plants
- (B) large and heavy so they stay in one place
- (C) alive only inside water
- (D) tiny, very many, and light enough to be carried far by wind

80. The arithmetic mean of the first five whole numbers 0, 1, 2, 3 and 4 is:

- (A) 2.5
- (B) 2
- (C) 4
- (D) 10

81. Why is the fuse always connected in the live wire and never in the path of a coloured 'earth' wire?

- (A) so that when it melts it cuts off the dangerous current to the appliance
- (B) to make the bulb glow brighter
- (C) because the earth wire carries the most current
- (D) so the appliance runs faster

82. A creeper grows the same length, x cm, each day. In 6 days it has grown 48 cm. Using $6x = 48$, the daily growth is:

- (A) 288 cm
- (B) 54 cm
- (C) 42 cm
- (D) 8 cm

83. A seed has a tough outer coat. Its main job is to:

- (A) protect the tiny embryo inside until conditions are right to grow
- (B) attract insects with bright colour
- (C) make the seed taste sweet
- (D) stop the seed from ever growing

84. A frequency table records how often each value appears, often using groups of marks like ||||.

These groups of marks are called:

- (A) pictographs
- (B) tally marks
- (C) medians
- (D) bar graphs

85. A fuse wire is made of a material with a LOW melting point so that it:

- (A) never melts however large the current
- (B) carries more current than a thick wire
- (C) glows brightly like a bulb filament
- (D) melts quickly when the current becomes too large

86. Solve the equation $3x = 2x + 8$, in which the unknown appears on both sides.

- (A) 8
- (B) 2
- (C) 4
- (D) 16

87. Wind, water and insects that carry pollen from one flower to another are together called:

- (A) agents of digestion
- (B) agents of pollination
- (C) decomposers
- (D) predators

88. A bar graph shows the number of seeds dispersed by four trees: P = 30, Q = 45, R = 25 and S = 50. The tree that dispersed the MOST seeds is:

- (A) R
- (B) P
- (C) S
- (D) Q

89. Two identical cells are joined so that the positive terminal of one touches the negative terminal of the other. This proper joining means their pushes:

- (A) turn into a magnetic field instead
- (B) stop any current from flowing
- (C) add up to give a larger total push
- (D) cancel out to give zero push

90. 'The sum of a number and 13 is exactly 13.' Written as $x + 13 = 13$, the number is:

- (A) 0
- (B) 13
- (C) 1
- (D) 26

91. When pollen from a flower lands on the stigma of the SAME flower (or another flower on the same plant), it is called:

- (A) fertilisation
- (B) dispersal
- (C) cross-pollination
- (D) self-pollination

92. Across a week a home used 6, 7, 5, 8, 6, 9 and 8 units of electricity. The range of the daily usage is:

- (A) 14
- (B) 4
- (C) 49
- (D) 9

93. An electric fan and a washing-machine motor spin because an electric current can produce a:

- (A) chemical effect that makes new gases
- (B) magnetic effect that makes a coil turn
- (C) cooling effect that freezes the axle
- (D) heating effect that boils the coil

94. Find the value of x in the equation $9 = x - 6$.

- (A) 54
- (B) 3
- (C) 6
- (D) 15

95. Grafting and layering, used by gardeners to make new fruit and flower plants, are examples of:

- (A) seed dispersal
- (B) sexual reproduction
- (C) artificial vegetative propagation
- (D) pollination

96. In the data set 8, 2, 8, 5, 2, 8, the value that appears most often (the mode) is:

- (A) 5
- (B) 33
- (C) 2
- (D) 8

97. Compact fluorescent lamps (CFLs) and LED bulbs are preferred over ordinary filament bulbs mainly because they:

- (A) need no electricity at all to glow
- (B) give the same light while using much less electricity
- (C) only work without any cell or supply
- (D) are much hotter and waste more energy

98. A number multiplied by 7 gives 0. Writing this as $7x = 0$, the value of the number is:

- (A) 70
- (B) 0
- (C) 7
- (D) 1

99. After fertilisation, the ovules inside the ovary of a flower develop into:

- (A) roots
- (B) seeds
- (C) petals
- (D) leaves

100. The mean (average) of two numbers is 7. If one of the numbers is 5, the other number is:

- (A) 2
- (B) 7
- (C) 9
- (D) 12